THE IDEAL CUTTING OUT EXACTLY THOSE BASE CHANGES ALONG WHICH A SURJECTION OF MODULES BECOMES AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- A projective module is a direct summand of a free module. Elementary standard fact used at proof lines 6659-6661, where a projective C with M direct sum C free is chosen; the counted proof cites no label. Allows the surjection u to be replaced by u direct sum the identity on C, so that M may be assumed free, which is the first reduction of the proof.
- Right exactness of the tensor product and the kernel of a base changed surjection. Elementary standard fact used at proof lines 6666-6668; the counted proof cites no label. Identifies Ker(u tensor 1) with the image of Ker(u) tensor_A B, which is precisely what makes the condition that the coefficients of Ker(u) vanish equivalent to u tensor 1 being injective.

Lemma 0.1. *Let A be a ring. Let $u : M \to N$ be a surjective map of A-modules. If M is projective as an A-module, then there exists an ideal $I \subset A$ such that for any ring map $\varphi : A \to B$ the following are equivalent*

(1) $u \otimes 1 : M \otimes_A B \to N \otimes_A B$ *is an isomorphism, and*
(2) $\varphi(I) = 0$.

Proof. As M is projective we can find a projective A-module C such that $F = M \oplus C$ is a free A-module. By replacing u by $u \oplus 1 : F = M \oplus C \to N \oplus C$ we see that we may assume M is free. In this case let I be the ideal of A generated by coefficients of all the elements of $\mathrm{Ker}(u)$ with respect to some (fixed) basis of M. The reason this works is that, since u is surjective and $\otimes_A B$ is right exact, $\mathrm{Ker}(u \otimes 1)$ is the image of $\mathrm{Ker}(u) \otimes_A B$ in $M \otimes_A B$. □

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